In blockchain applications, all the overt logic is outside, with the off-chain data and functions. The blockchain is like a covert observer that systematically records relevant information on-chain about activities that happen off-chain.

You may wonder why, then, you didn’t write thousands of lines of code and multiple classes for the smart contract code. Even though smart contracts are simple and succinct, a lot of operations are carried out by the blockchain infrastructure behind the scenes. This situation is similar to the famous MapReduce algorithm used in big data analytics that fits within a single page. In that case, all the work is done behind the scenes by the MapReduce infrastructure.

6.6 Best practices

Here are some best practices, specifically focusing on on-chain and off-chain data:

- Blockchain program is not about translating or porting an application in a traditional programming language (say, Java) into a blockchain programming language such as Solidity. Define in a smart contract only the data needed for on-chain recording--nothing more. In the ASK Dapp, for example, only two data structures are defined for the request data and response data that needs to be recorded.